DA4 – Assessment Question

Subject: Application Development and Deployment Architecture

22MIC0020

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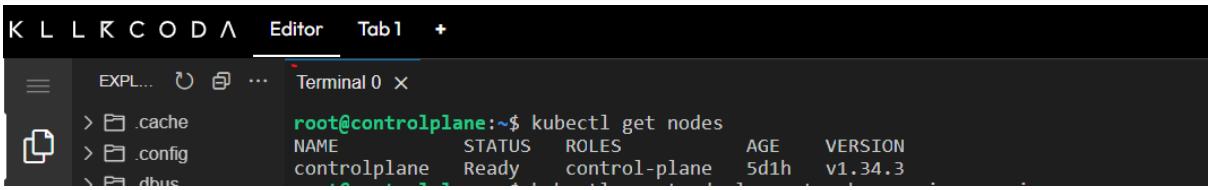
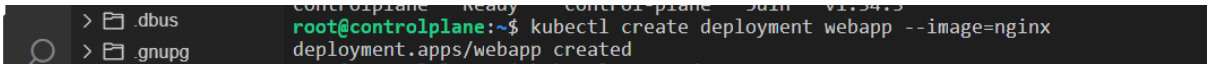
Technical Stack/Framework: Kubernetes and CGroup

A DevOps engineer needs to deploy and manage a containerized web application using **Kubernetes** in the **KillerCoda Kubernetes playground**. The engineer must also ensure that the application can scale when traffic increases and that system resources such as CPU and memory are controlled using **CGroup**.

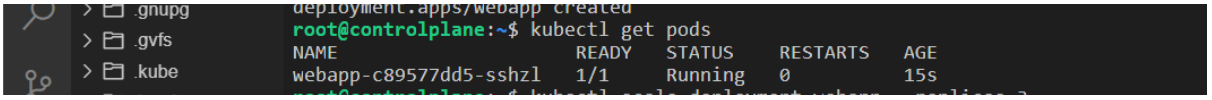
1. Access the Kubernetes playground and verify that the Kubernetes cluster is running by listing the available nodes.



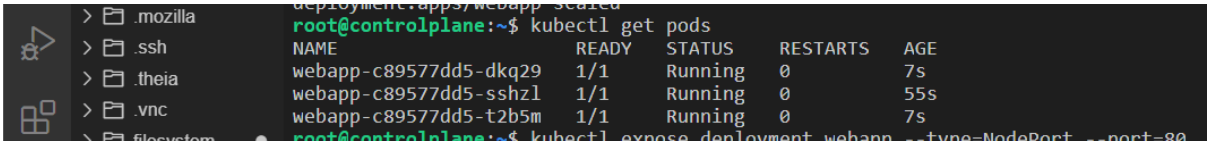
2. Create a deployment named **webapp** using the **nginx container image**.



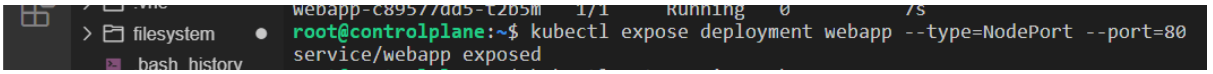
3. Verify that the pod created by the deployment is running successfully.

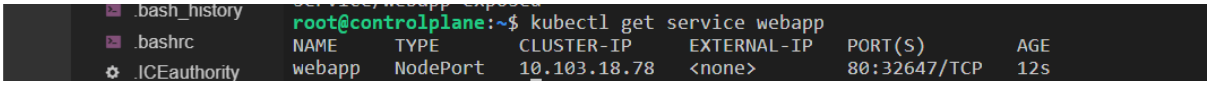


4. Scale the deployment to **3 replicas** and confirm that additional pods are created automatically.

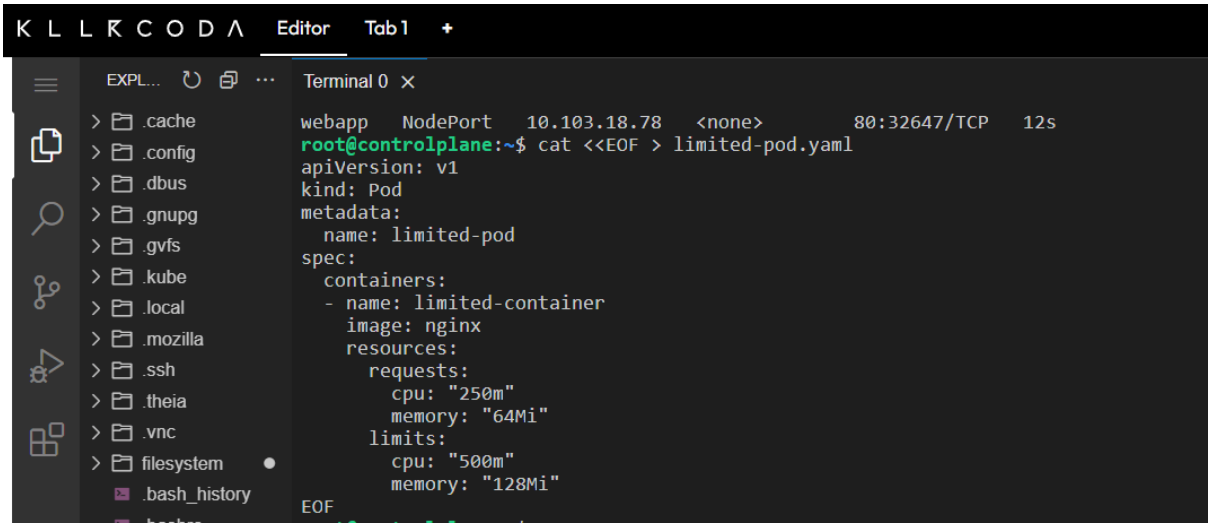


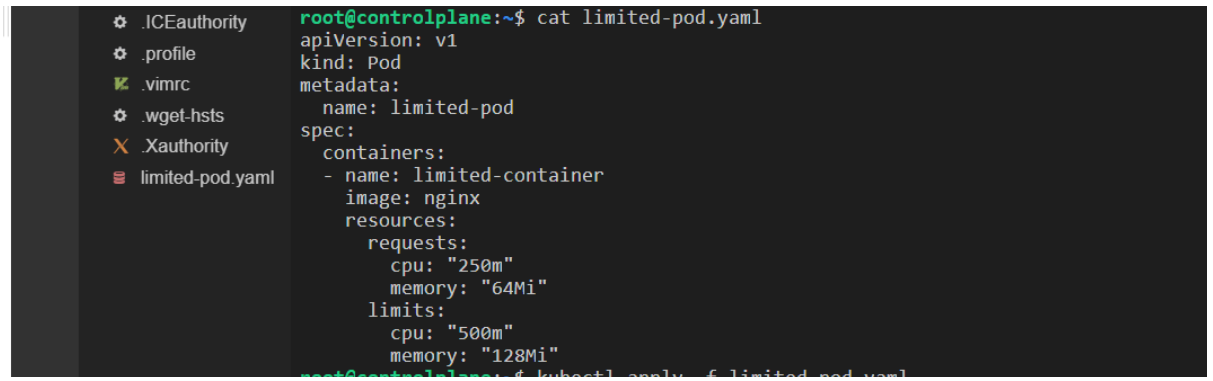
5. Expose the deployment using a **NodePort service** and display the service details.





6. Create a YAML configuration file to deploy a pod named **limited-pod** with resource limits (**CPU: 500m and Memory: 128Mi**).





7. Apply the configuration file and verify that the pod is created successfully.

```
root@controlplane:~$ kubectl apply -f limited-pod.yaml
pod/limited-pod created
root@controlplane:~$ kubectl get pod limited-pod
NAME          READY   STATUS    RESTARTS   AGE
limited-pod   1/1     Running   0          11s
root@controlplane:~$ kubectl describe pod limited-pod
Name:             limited-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             controlplane/172.30.1.2
Start Time:       Mon, 09 Mar 2026 10:53:20 +0000
Labels:           <none>
Annotations:      cni.projectcalico.org/containerID: bf155376424e61432e96639ad9224307df4663be789712dc45b513b3489e98c6
                  cni.projectcalico.org/podIP: 192.168.0.9/32
                  cni.projectcalico.org/podIPs: 192.168.0.9/32
Status:           Running
IP:               192.168.0.9
IPs:
  IP:  192.168.0.9
Containers:
  limited-container:
    Container ID:   containerd://8541d44504db192d38fd6f874a7e3220200b965b437750fc57a969102a9c0361
    Image:          nginx
    Image ID:       docker.io/library/nginx@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Mon, 09 Mar 2026 10:53:21 +0000
    Ready:          True
    Restart Count:  0
    Limits:
      cpu:     500m
      memory:  128Mi
    Requests:
      cpu:        250m
      memory:     64Mi
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-t9fbn (ro)
```

8. Describe the pod and confirm that the resource limits are enforced using **CGroup**.

```
KLLRCODA  Editor  Tab1  +
EXPL...
.cache
.config
.dbus
.gnupg
.gvfs
.kube
.local
.mozilla
.ssh
.theia
.vnc
filesystem
.bash_history
.bashrc
.ICEauthority
.profile
.vimrc
.wget-hsts
.Xauthority
limited-pod.yaml
Terminal 0
root@controlplane:~$ kubectl describe pod limited-pod
Name:             limited-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             controlplane/172.30.1.2
Start Time:       Mon, 09 Mar 2026 10:53:20 +0000
Labels:           <none>
Annotations:      cni.projectcalico.org/containerID: bf155376424e61432e96639ad9224307df4663be789712dc45b513b3489e98c6
                  cni.projectcalico.org/podIP: 192.168.0.9/32
                  cni.projectcalico.org/podIPs: 192.168.0.9/32
Status:           Running
IP:               192.168.0.9
IPs:
  IP:  192.168.0.9
Containers:
  limited-container:
    Container ID:   containerd://8541d44504db192d38fd6f874a7e3220200b965b437750fc57a969102a9c0361
    Image:          nginx
    Image ID:       docker.io/library/nginx@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Mon, 09 Mar 2026 10:53:21 +0000
    Ready:          True
    Restart Count:  0
    Limits:
      cpu:     500m
      memory:  128Mi
    Requests:
      cpu:        250m
      memory:     64Mi
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-t9fbn (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
```

```
QoS Class:                   Burstable
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  22s   default-scheduler  Successfully assigned default/limited-pod to controlplane
  Normal  Pulling    21s   kubelet            Pulling image "nginx"
  Normal  Pulled     21s   kubelet            Successfully pulled image "nginx" in 137ms (137ms including waiting). Image size: 62944796 bytes.
  Normal  Created    21s   kubelet            Created container: limited-container
  Normal  Started    21s   kubelet            Started container limited-container
root@controlplane:~$
```

```
root@controlplane:~$ kubectl get pod limited-pod -o wide
NAME          READY   STATUS    RESTARTS   AGE     IP            NODE           NOMINATED NODE   READINESS GATES
limited-pod   1/1     Running   0          4m57s   192.168.0.9   controlplane   <none>           <none>
```

```
root@controlplane:~$ ssh 172.30.1.2
The authenticity of host '172.30.1.2 (172.30.1.2)' can't be established.
ED25519 key fingerprint is SHA256:XikjTSvie4xfSHl8CD14UKeq6b3m5zuJAlVVC6JLlao.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:2: [hashed name]
    ~/.ssh/known_hosts:3: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? Y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '172.30.1.2' (ED25519) to the list of known hosts.
ssh_dispatch_run_fatal: Connection to 172.30.1.2 port 22: Broken pipe
root@controlplane:~$
```

9. Display all Kubernetes resources created during the experiment.

```
root@controlplane:~$ kubectl get all
NAME                              READY   STATUS    RESTARTS   AGE
pod/limited-pod                   1/1     Running   0          18m
pod/webapp-c89577dd5-dkq29        1/1     Running   0          22m
pod/webapp-c89577dd5-sshzl        1/1     Running   0          23m
pod/webapp-c89577dd5-t2b5m        1/1     Running   0          22m

NAME                 TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
service/kubernetes   ClusterIP   10.96.0.1      <none>        443/TCP        5d1h
service/webapp       NodePort    10.103.18.78   <none>        80:32647/TCP   22m

NAME                     READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/webapp   3/3     3            3           23m

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/webapp-c89577dd5    3         3         3       23m
root@controlplane:~$
```

KLLKCODA Editor Tab 1 +

EXPL...

- .cache
- .config
- .dbus
- .gnupg
- .gvfs
- .kube
- .local
- .mozilla
- .ssh
- .theia
- .vnc
- filesystem
- .bash_history
- .bashrc
- .ICEauthority
- .profile
- .vimrc
- .wget-hsts
- .Xauthority
- limited-pod.yaml

Terminal 0

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Name:             limited-pod
Namespace:        default
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Service Account:  default
Node:             controlplane/172.30.1.2
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                  cni.projectcalico.org/podIPs: 192.168.0.9/32
Status:           Running
IP:               192.168.0.9
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  IP:  192.168.0.9
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    Image ID:       docker.io/library/nginx@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Mon, 09 Mar 2026 10:53:21 +0000
    Ready:          True
    Restart Count:  0
    Limits:
      cpu:     500m
      memory:  128Mi
    Requests:
      cpu:        250m
      memory:     64Mi
    Environment:  <none>
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      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-t9fbn (ro)
Conditions:
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  PodReadyToStartContainers   True
  Initialized                 True
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  Normal  Pulled     21s   kubelet            Successfully pulled image "nginx" in 137ms (137ms including waiting). Image size: 62944796 bytes.
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  Normal  Started    21s   kubelet            Started container limited-container
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root@controlplane:~$ kubectl get pod limited-pod -o wide
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root@controlplane:~$ ssh 172.30.1.2
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    ~/.ssh/known_hosts:2: [hashed name]
    ~/.ssh/known_hosts:3: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? Y
Please type 'yes', 'no' or the fingerprint: yes
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